

Indian Statistical Institute
M. Math 2nd year
Academic year 2025-2026
Fourth semester
Midterm Examination
Course: Teichmüller theory
23.25 - 02 - 26
2 hours

- Answer as many questions as you can.
- You may use results proved in class, but make sure to state them clearly.
- Maximum marks is 40.
- Unless otherwise mentioned, all Riemann surfaces are assumed to be connected.

1. For $T \in GL(n, \mathbb{R})$, let

$$M(T) := \sup_{\|v\|=1} \|T(v)\|, m(T) := \inf_{\|v\|=1} \|T(v)\|, K(T) := M(T)/m(T).$$

- a) Show that $K(ST) \leq K(S)K(T)$ for all S, T .
- b) Show that $K(T) = 1$ if and only if $\|T\|^n = |\det T|$.
- c) If $n = 2$, then show that $\|T\|^2 = K(T)|\det T|$. (3+4+4 = 10 marks)

2. Let $G = SL(2, \mathbb{R})$, $K = SO(2, \mathbb{R})$. For $g \in G$, let $\mu_g \in \mathbb{D}$ denote the Beltrami coefficient of g .

- a) Show that the map $G \rightarrow \mathbb{D}, g \mapsto \mu_{g^{-1}}$ induces a bijection $\Phi : G/K \rightarrow \mathbb{D}$.
- b) The action of G on G/K induces via the bijection Φ an action of G on $\mathbb{D}$. Show that for any $g \in G$, the map $\phi_g : \mathbb{D} \rightarrow \mathbb{D}, \mu \mapsto g \cdot \mu$ is a Möbius transformation of $\mathbb{D}$. (4+8 = 12 marks)

3. Let f, g be K -QC maps defined on $U \subset \mathbb{C}$. Show that $\mu_f = \mu_g$ a.e. if and only if $g = \phi \circ f$ for some conformal map ϕ . (8 marks)
4. Let $A_1, A_2 \subset \mathbb{C}$ be annuli. Show that if $A_1 \subset A_2$ then $m(A_1) \leq m(A_2)$ (here $m(A)$ denotes the modulus of an annulus A). (5 marks)
5. a) Show that there does not exist a QC map $f : \mathbb{C}^* \rightarrow \mathbb{D}^*$.
b) Show that there does not exist a QC map $f : \mathbb{C} \rightarrow \mathbb{D}$.
c) Show that if there is a QC map $f : X \rightarrow Y$ between Riemann surfaces X, Y , then the universal covers of X, Y are biholomorphic. (7+3+5 = 15 marks)